CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Düsseldorf • EU Blue Card linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in knowledge graphs for AI systems, from extracting entities and relationship types out of complex documents and having domain experts validate graph changes to making the graph usable for AI agents and retrieval. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by study work on ontologies with Protégé, RDF, OWL and SPARQL and consulting with business stakeholders at Covestro.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer of the knowledge graph behind an agentic knowledge-extraction system for the German energy sector: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) that turn complex regulatory and technical documents into machine-readable graph entities and relationship types, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, letting domain experts validate and approve knowledge-graph changes.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph to AI agents for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Used Claude Code, custom skills and agentic coding workflows to prototype fast and ship reliable code grounded in clean code principles.
- **Tech stack: Neo4j, GraphRAG, Python, LangGraph, LangChain, LiteLLM, MCP, Postgres, FastAPI, React, Docker, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows and translated their requirements into the solution approach; designed and deployed an agentic LangGraph application that routes each request to vector database queries, general inference or summarization.
- Developed a React frontend and a Python backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face, PyTorch, Vector Databases, AWS CDK**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) for the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time series analysis of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Knowledge Graph | NLP (NLTK) | Protégé | SPARQL | RDF/OWL

- Knowledge graph as a learning resource for AI students (Protégé, rdflib): NLTK information extraction, RDF transformation, SPARQL querying, OWL enhancement.

Project: Semantic Analytics Agent | OWL | SKOS | SPARQL | Triple Store | LangGraph

- Modeled OWL ontologies and SKOS taxonomies for synthetic life sciences (pharma) commercial data with entity hierarchies, inference rules and SHACL governance checks in a Stardog triple store, kept separate from the semantic data model; a LangGraph agent serves conversational analytics only through approved SPARQL queries, tested with a golden-question set in CI.

TECHNICAL SKILLSET

- **Knowledge Graphs:** graph databases (Neo4j), GraphRAG, ontologies (Protégé, OWL), RDF, SPARQL, rdflib, entity and relationship extraction
- **AI & NLP:** LangGraph, LangChain, LiteLLM, RAG, MCP, prompt engineering, NLTK, Hugging Face, RAGAS
- **Engineering:** Python, SQL, FastAPI, Postgres, MongoDB, Docker, Kubernetes, AWS, Azure, GitLab CI/CD
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India